$$\alpha = \frac{\text{Number of moles dissociated}}{\text{total number of moles}}$$

We shall derive an expression for ostwald's law by considering a weak acid, i.e. acetic acid (CH_3COOH). The dissociation of acetic acid can be represented as

$$CH_3COOH \rightleftharpoons H^+ + CH_3COO^-$$

The dissociation constant of acetic acid is,

$$k_a = \frac{[H^+][CH_3COO^-]}{[CH_3COOH]} \qquad .....(8.13)$$

	CH_3COOH	H^+	CH_3COO^-
Initial number of moles	1	-	-
Degree of dissociation of CH_3COOH	α	-	-
Number of moles at equilibrium	$1-\alpha$	α	α
Equilibrium concentration	$(1-\alpha)\,C$	$\alpha\,C$	$\alpha\,C$

Substituting the equilibrium concentration in equation (8.13)

$$K_a = \frac{(\alpha C)(\alpha C)}{(1-\alpha)C}$$

$$K_a = \frac{\alpha^2 C}{1-\alpha} \qquad .....(8.14)$$

We know that weak acid dissociates only to a very small extent. Compared to one, α is so small and hence in the denominator $(1-\alpha) \simeq 1$. The above expression (8.14) now becomes,

$$K_a = \alpha^2 C$$

$$\Rightarrow \alpha^2 = \frac{K_a}{C}$$

$$\alpha = \sqrt{\frac{K_a}{C}} \qquad .....(8.15)$$

Let us consider an acid with K_a value 4×10^{-4} and calculate the degree of dissociation of that acid at two different concentration 1×10^{-2}M and 1×10^{-4}M using the above expression (8.15)

For 1×10^{-2}M,

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